

## BEKTAU-ATA, Kazakhstan

WMO: 356990

Lat: 47.450N

Lon: 74.817E

Elev: 2034

StdP: 13.65

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |             | Humidification DP/MCDB and HR |            |             |              |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|--------------|-------------|-------------------------------|------------|-------------|--------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|--------------|
|               |              |             | 99.6%                         |            |             | 99%          |            |             | 0.4%                  |             | 1%          |             |                       |           |              |
|               | 99.6%        | 99%         | DP                            | HR         | MCDB        | DP           | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-11.2 | (c)<br>-7.2 | (d)<br>-19.7                  | (e)<br>2.0 | (f)<br>-8.8 | (g)<br>-16.0 | (h)<br>2.5 | (i)<br>-4.2 | (j)<br>34.0           | (k)<br>24.9 | (l)<br>27.1 | (m)<br>10.1 | (n)<br>8.0            | (o)<br>20 | (p)<br>0.518 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>18.6            | (c)<br>91.5     | (d)<br>63.9 | (e)<br>88.5 | (f)<br>62.8 | (g)<br>85.7 | (h)<br>61.9 | (i)<br>66.0         | (j)<br>85.6 | (k)<br>64.8 | (l)<br>83.9 | (m)<br>63.7 | (n)<br>81.9 | (o)<br>4.4           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 59.7                            | 82.5 | 71.9 | 58.0 | 77.4 | 71.7 | 56.3 | 72.8 | 71.7 | 31.9          | 85.8 | 31.0 | 84.1 | 30.1 | 81.9 | 73.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 22.4              | 20.4 | 16.9 | -16.5                      | 96.4 | 3.4                | 3.0 | -19.0                                              | 98.6 | -21.0      | 100.3 | -22.9      | 102.0 | -25.4      | 104.1 |
|                   |      |      | -16.9                      | 69.5 | 3.3                | 1.9 | -19.3                                              | 70.9 | -21.3      | 72.0  | -23.1      | 73.0  | -25.6      | 74.4  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.0   | 10.1  | 12.7  | 29.3  | 49.4  | 60.4  | 70.5  | 72.9  | 70.8  | 60.0  | 45.4  | 28.6  | 15.5  |  |  |
|                                                                        | DBStd       | 24.47  | 9.23  | 10.07 | 11.00 | 10.09 | 8.06  | 6.87  | 6.59  | 7.07  | 9.03  | 9.46  | 10.71 | 10.33 |  |  |
|                                                                        | HDD50       | 5001   | 1236  | 1045  | 644   | 130   | 12    | 0     | 0     | 0     | 20    | 201   | 644   | 1069  |  |  |
|                                                                        | HDD65       | 8427   | 1701  | 1465  | 1105  | 472   | 184   | 29    | 12    | 27    | 199   | 607   | 1092  | 1534  |  |  |
|                                                                        | CDD50       | 2800   | 0     | 0     | 4     | 112   | 334   | 614   | 710   | 645   | 321   | 59    | 1     | 0     |  |  |
|                                                                        | CDD65       | 752    | 0     | 0     | 0     | 4     | 41    | 192   | 257   | 207   | 50    | 1     | 0     | 0     |  |  |
|                                                                        | CDH74       | 7158   | 0     | 0     | 0     | 33    | 348   | 1806  | 2486  | 2019  | 462   | 4     | 0     | 0     |  |  |
|                                                                        | CDH80       | 2479   | 0     | 0     | 0     | 5     | 65    | 595   | 930   | 752   | 132   | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 6.4    | 7.7   | 7.6   | 7.7   | 6.1   | 6.5   | 5.8   | 5.8   | 6.2   | 6.3   | 5.7   | 5.0   | 6.1   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 8.3    | 0.7   | 0.6   | 0.6   | 0.6   | 0.9   | 0.8   | 1.1   | 0.6   | 0.2   | 0.5   | 0.8   | 0.7   |  |  |
|                                                                        | PrecMax     | 13.4   | 1.3   | 1.3   | 1.5   | 1.4   | 2.9   | 3.8   | 3.2   | 1.5   | 0.6   | 1.4   | 2.1   | 2.8   |  |  |
|                                                                        | PrecMin     | 5.2    | 0.2   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   |  |  |
|                                                                        | PrecStd     | 1.9    | 0.4   | 0.3   | 0.4   | 0.4   | 0.6   | 0.7   | 0.9   | 0.4   | 0.2   | 0.4   | 0.5   | 0.5   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 33.1  | 39.2  | 62.7  | 78.3  | 85.8  | 91.8  | 97.3  | 95.0  | 89.3  | 73.1  | 57.7  | 39.5  |  |  |
|                                                                        |             | MCWB   | 30.4  | 35.1  | 46.4  | 55.3  | 58.9  | 64.5  | 65.8  | 64.7  | 59.9  | 52.7  | 45.6  | 32.2  |  |  |
|                                                                        | 2%          | DB     | 29.9  | 33.4  | 56.2  | 72.9  | 81.2  | 88.9  | 92.1  | 90.9  | 83.6  | 68.1  | 51.5  | 34.3  |  |  |
|                                                                        |             | MCWB   | 27.7  | 30.0  | 43.7  | 53.6  | 58.3  | 63.7  | 64.6  | 63.1  | 58.2  | 50.1  | 44.1  | 30.5  |  |  |
|                                                                        | 5%          | DB     | 26.1  | 30.2  | 49.8  | 69.2  | 77.7  | 86.2  | 88.4  | 87.5  | 79.3  | 64.1  | 46.4  | 31.4  |  |  |
|                                                                        |             | MCWB   | 23.7  | 27.6  | 41.2  | 52.0  | 57.1  | 62.7  | 63.3  | 61.9  | 56.3  | 48.1  | 39.9  | 28.6  |  |  |
|                                                                        | 10%         | DB     | 22.9  | 27.0  | 44.6  | 65.2  | 74.6  | 83.5  | 85.3  | 84.2  | 75.0  | 59.9  | 42.4  | 28.7  |  |  |
|                                                                        |             | MCWB   | 21.0  | 24.8  | 38.4  | 50.1  | 55.8  | 61.7  | 62.4  | 61.0  | 55.1  | 46.3  | 36.7  | 26.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 31.2  | 34.5  | 49.3  | 57.6  | 63.1  | 67.5  | 68.0  | 66.9  | 62.1  | 55.6  | 49.9  | 34.0  |  |  |
|                                                                        |             | MCDB   | 32.3  | 38.1  | 59.0  | 71.3  | 79.8  | 85.4  | 87.3  | 88.5  | 85.6  | 67.2  | 54.9  | 36.1  |  |  |
|                                                                        | 2%          | WB     | 27.6  | 30.8  | 45.2  | 54.9  | 60.5  | 65.6  | 66.3  | 64.9  | 59.5  | 52.0  | 45.0  | 31.0  |  |  |
|                                                                        |             | MCDB   | 29.7  | 33.1  | 54.5  | 69.7  | 76.1  | 84.2  | 85.8  | 85.9  | 79.6  | 65.3  | 49.6  | 33.7  |  |  |
|                                                                        | 5%          | WB     | 24.2  | 27.8  | 41.5  | 53.4  | 58.7  | 64.3  | 65.2  | 63.5  | 57.5  | 49.6  | 40.8  | 28.8  |  |  |
|                                                                        |             | MCDB   | 25.8  | 29.9  | 48.4  | 67.3  | 73.7  | 82.8  | 83.9  | 83.6  | 76.7  | 61.7  | 46.1  | 31.3  |  |  |
|                                                                        | 10%         | WB     | 20.9  | 24.8  | 38.5  | 51.3  | 57.1  | 62.9  | 64.1  | 62.2  | 55.8  | 47.2  | 37.2  | 26.3  |  |  |
|                                                                        |             | MCDB   | 22.9  | 27.0  | 44.5  | 63.4  | 72.6  | 80.5  | 81.8  | 81.0  | 73.1  | 57.9  | 42.0  | 28.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.4  | 12.7  | 13.7  | 18.5  | 20.0  | 19.2  | 18.6  | 19.9  | 19.7  | 17.2  | 13.1  | 11.5  |  |  |
|                                                                        |             | MCDBR  | 12.5  | 14.3  | 19.7  | 22.4  | 22.4  | 21.2  | 21.1  | 22.9  | 24.1  | 21.7  | 16.7  | 11.9  |  |  |
|                                                                        |             | MCWBR  | 10.6  | 12.1  | 12.5  | 11.3  | 10.5  | 8.8   | 8.2   | 9.0   | 10.5  | 11.5  | 10.7  | 9.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 11.2  | 14.1  | 18.4  | 20.1  | 20.9  | 19.7  | 19.4  | 20.9  | 22.7  | 19.8  | 15.7  | 11.1  |  |  |
|                                                                        |             | MCWBR  | 10.0  | 12.4  | 12.4  | 10.9  | 10.6  | 8.5   | 8.2   | 8.8   | 10.8  | 11.7  | 11.1  | 9.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.297 | 0.327 | 0.377 | 0.417 | 0.422 | 0.418 | 0.432 | 0.386 | 0.344 | 0.344 | 0.328 | 0.300 |  |  |
|                                                                        | taud        |        | 2.077 | 2.013 | 2.059 | 2.085 | 2.123 | 2.204 | 2.176 | 2.294 | 2.371 | 2.283 | 2.200 | 2.122 |  |  |
|                                                                        | Ebn at Noon |        | 237   | 256   | 262   | 264   | 267   | 269   | 263   | 271   | 274   | 251   | 226   | 220   |  |  |
|                                                                        | Edh at Noon |        | 29    | 39    | 44    | 48    | 48    | 44    | 45    | 38    | 32    | 30    | 26    | 25    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 539   | 880   | 1269  | 1720  | 2115  | 2215  | 2092  | 1968  | 1536  | 938   | 521   | 382   |  |  |
|                                                                        | RadStd      |        | 61    | 66    | 95    | 104   | 114   | 125   | 149   | 96    | 81    | 74    | 50    | 53    |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | +0.94 | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (3 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page